

SpaceLoop: Complete Features & Architectural Specification

A Mid-Detailed Engineering Dossier on AI Capabilities, Zero-Hardware Protocols & Micro-Leasing

Project: SpaceLoop (Built for Hack2Ignite 2026)	Version: 2.4 (Production Architecture)	Stack: Python / Flask / SQLite / Dual LLMs
Lead Dev: Kanishk Singh	Compliance: DPDP Act 2023 & India Stack	Deployment: Live Cloud + Render Blueprint

1. Executive Overview & The Problem Solved

Modern metropolitan areas suffer from a pronounced real estate inefficiency paradox: hundreds of millions of square feet sit vacant or severely underutilized during daytime and off-peak hours (empty residential garages, daytime-vacant basements, retail storefronts idle after 4 PM, private driveways during working hours, and off-peak cafés). Simultaneously, students, digital creators, remote freelancers, and micro-entrepreneurs face prohibitive barriers—rigid 12-month commercial leases, \$400+/day production studios, and multi-month security deposits. SpaceLoop solves this disparity by converting idle urban square footage into liquid, on-demand, temporary micro-rentals (by the hour or day) backed by automated legal safeguards, multimodal spatial inspection, and zero-hardware access protocols.

Architectural Core: High-Availability Dual-AI Engine

Multi-Tier Intelligence Hierarchy: SpaceLoop deploys a three-tier resilient AI framework. High-complexity spatial inspections and conversational search queries route first to **Groq (Llama-3.3-70B-Versatile)** for sub-second responses (<450ms). If Groq experiences rate limits or network issues, requests automatically fall over to **Google Gemini 1.5 Flash**. In offline, air-gapped, or degraded connectivity scenarios, a deterministic rule-based heuristic engine guarantees 100% platform uptime.

Key Market Dynamics & Operational Comparison

Platform / Model	Target Use Case	Hardware & Capital Barrier	Legal Tenancy Risk
Airbnb / Vrbo	Multi-day overnight hospitality	High host furnishing & smart lock CapEx	High risk of overstay & squatting
Commercial Leases	Multi-year office / retail space	Predatory multi-month security deposits	Rigid statutory tenancy leasehold
SpaceLoop (Ours)	On-demand micro-utility (1–8 hrs)	■0 Hardware (GPS + Laminated QR)	Revocable License (Zero Tenancy)

2. Master Subsystems & Feature Breakdown Matrix

SpaceLoop comprises 12 interconnected subsystems spanning computer vision, legal contract automation, fintech escrow, and physical access verification. The matrix below outlines each capability, its technical implementation, and operational impact:

Subsystem / Feature	Domain	Core Technology	Operational Function & Impact
1. Multimodal Space Inspector	Spatial AI	Groq Llama 3.3 70B / Gemini 1.5	Extracts sqft, lighting, acoustic dB (<36dB), circuit load, and auto-generates high-converting listing copy.
2. Conversational Matchmaker	NLP Search	Semantic Intent Parser + Scorer	Parses conversational renter queries; outputs a 0–100% Compatibility Score with pros/cons breakdown.
3. AI Micro-Lease Synthesizer	Legal Tech	LLM Contract Engine (Sec 52)	Synthesizes enforceable Revocable License Agreements; prevents tenancy rights and caps host liability.
4. Dynamic Pricing Calculator	Economics	SqFt Polynomial Multiplier Model	Computes hourly/daily pricing baselines; projects monthly passive income and commercial lease savings.
5. Zero-Hardware Access Suite	Physical Access	Haversine GPS + 4-Digit Handshake	Enables physical access verification via laminated door QRs and rotating PINs without smart lock costs.
6. CV Room Condition Delta	Computer Vision	Visual Diff + Appliance Detection	Compares entry vs exit photos; verifies fan/light shutdown and auto-approves instant UPI escrow release.
7. Student Academic Verification	India Stack	Domain Regex (.ac.in) + ID Masking	Validates enrolled student credentials for subsidized study rates while masking sensitive identity data.
8. DigiLocker Aadhaar OTP	KYC & DPDP	DPDP Salted SHA-256 Hash Token	Zero raw Aadhaar storage; masks display to XXXX-XXXX-1234 and secures user privacy under DPDP Act 2023.
9. Host Electricity Bill (Discom)	Property Proof	BBPS / Discom CA Number Lookup	Cross-verifies utility consumer accounts (BESCOM, TPDDL) to confirm legal possession of space.
10. NPCI UPI ■1 Penny Drop	FinTech KYC	NPCI UPI VPA Resolution API	Validates host bank account title via penny drop before payout activation to prevent fraud.
11. Objective Trust Index (OTI)	Reputation	Multi-Variate Telemetry Scoring	Replaces biased 5-star reviews; weights punctuality (35%), cleanliness (35%), KYC (20%), and disputes (10%).
12. LoopBot AI Concierge	AI Assistant	Context-Aware Multi-Turn LLM Bot	Assists seekers with space recommendations, helps hosts optimize rates, and pre-answers inquiries.

End-to-End Platform Lifecycle

End-to-End User Journey: Host uploads space photo → AI Inspector calculates specs & sets baseline dynamic rate → Listing goes live → Renter searches in natural language → AI Matchmaker ranks spaces with compatibility score → Renter books space → AI synthesizes customized Micro-Lease → Renter arrives and checks in via GPS & Door QR → Renter completes session and submits checkout scan → AI evaluates condition delta → Instant ■100 UPI escrow deposit refund.

3. Technical Deep-Dive: Spatial AI, Discovery & Micro-Leasing

Feature 1: Multimodal AI Space Inspector

Challenge: Manual listing creation is tedious and inaccurate. Property hosts rarely know exact square footage, electrical circuit loads, or how to craft high-converting marketing descriptions.

Architecture: When a host uploads property imagery and unstructured notes (e.g. *'quiet 2-car garage with white walls and lots of plugs'*), the AI Space Inspector executes a structured multi-phase analysis:

- **Spatial Metric Estimation:** Estimates usable floor area (30–25,000 sq ft) and maximum occupancy capacity.
- **Environmental Profiling:** Categorizes natural vs artificial illumination (e.g. *'Bright Natural Sunlit with Sheer Diffusers'*) and background acoustics (e.g. *'Quiet (<38 dB)'*).
- **Power Infrastructure:** Detects electrical outlets and circuits (e.g. *'4x Grounded 20A Circuits + Level 2 EV Charger'*).
- **Security Guardrails:** Host notes are isolated within strict XML tags (`<user_untrusted_notes>`) to prevent prompt injection and system instruction leaks.

Feature 2: Natural Language Semantic Matchmaker

Challenge: Dropdown filters fail when seekers have nuanced, activity-specific criteria.

Architecture: Seekers type conversational search queries such as: *'Need an acoustically quiet studio for a 3-person podcast recording this Saturday under \$40/hr with good Wi-Fi.'*

- The semantic parser extracts intent vectors: domain (Podcast), capacity (3 pax), acoustic needs (quiet), budget (\$40/hr), and essential amenities.
- Evaluates candidate spaces in real time and computes a **0–100% Compatibility Score**.
- Returns diagnostic badges (*'Top Pick'*, *'Great Match'*, *'Alternative'*), specific pros/cons, and budget compliance notes.
- Supported by a rule-based fallback keyword engine guaranteeing zero query failures even if external LLM APIs are offline.

Feature 3: Automated Plain-English AI Micro-Lease Synthesizer

Legal Innovation & Tenancy Prevention: The biggest concern for property owners is tenant squatting and legal lease lock-in. In India, long-term tenancy falls under the Transfer of Property Act 1882 (Section 105), which grants heavy tenant protections. SpaceLoop completely eliminates this friction by algorithmically generating an enforceable, plain-English **Temporary Space Use License Agreement** under **Section 52 of the Indian Easements Act, 1882**.

- **Revocable License Clause:** Explicitly stipulates that no tenancy or leasehold interest is created; access is strictly a revocable personal license.
- **Activity-Bound Scope:** Confines permission strictly to the declared activity (e.g. study session, video shoot), prohibiting subletting or unauthorized retail.
- **Liquidated Overstay Damages:** Automatically charges an overtime penalty of **1.5x the hourly rate** in 30-minute increments for late vacating.
- **Mutual Indemnification:** Shifts premises liability for personal belongings and minor injury onto the licensee, protecting the property owner.

Feature 4: Dynamic Micro-Pricing & Passive Revenue Calculator

Economics Engine: SpaceLoop deploys an automated non-linear pricing model designed to maximize host occupancy while providing 60–75% cost savings to renters:

- **Category Baseline Rates:** Storage (■35/hr), Studio (■65/hr), Parking (■25/hr), Pop-up/Retail (■95/hr), Event (■85/hr), Workspace (■45/hr).
- **Non-Linear SqFt Scaling:** Evaluates $\text{sqft_adj} = \max(0.8, \min(2.5, \text{sqft} / 250.0))$ with dampening formula $\text{Rate} = \text{Base} * (0.6 + 0.4 * \text{sqft_adj})$.
- **Yield Projections:** Calculates projected monthly net income assuming a realistic 12-day monthly occupancy, deducting the 15% platform service fee.

Features 5 – 8: Zero-Hardware Physical Access & Objective Trust

Feature 5: Zero-Hardware Smart Access Protocol

Why Zero-Hardware? Smart IoT locks cost \$150–\$300 per door, demand reliable 24/7 Wi-Fi, and fail during power outages. In developing urban environments, smart locks represent an impossible financial barrier for spare spaces. SpaceLoop introduces a zero-CapEx, four-layered access architecture:

Access Mechanism	Hardware Cost	Security & Verification Protocol
1. GPS Haversine Geofence	■0 (Smartphone GPS)	Calculates real-time distance using Haversine formula; restricts check-in strictly to within 30–50m radius of property coordinates.
2. Printable Cryptographic Door QR	■10 (Laminated Sheet)	Affixed to entrance door. Encodes a 64-character cryptographic SHA-256 room token verified upon camera scan.
3. Caretaker Handshake PIN	■0 (Verbal / WhatsApp)	Rotating 4-digit PIN generated per booking. Guest presents code to on-site guard or caretaker for physical gate entry.
4. Mechanical Keybox	■800 (One-time mechanical)	Weatherproof 4-wheel mechanical key lockbox. The master key is stored inside; combination code rotates per reservation.

Feature 6: Computer Vision Room Condition Delta Evaluation

Automated Inspection & Escrow Release: Hosts cannot physically inspect rooms after every 2-hour rental. SpaceLoop solves this with an automated multimodal Computer Vision pipeline at check-out:

- **Visual Diff Analysis:** The renter captures a photo scan of the room upon departure, which is compared against the check-in baseline.
- **Cleanliness & Waste Detection:** Flags abandoned garbage, drink cans, or misplaced furniture.
- **Appliance & Energy Conservation Check:** Confirms whether ceiling fans, lights, and air conditioners have been powered off.
- **Instant Escrow Refund:** Upon passing the AI inspection (Condition Match Score ≥ 90%), the system programmatically triggers the instant release of the ■100 UPI security deposit back to the renter's account.

Feature 7: India-First Trust & Document Verification Suite

To establish unquestioned institutional trust, SpaceLoop incorporates four integrated verification gateways:

- **7A. Student Academic Verification:** Validates academic credentials via institutional email domain handshakes (e.g. .ac.in, .edu.in) or student ID uploads, unlocking subsidized rates while masking IDs (e.g. STU-***-4821).
- **7B. DigiLocker Aadhaar OTP (DPDP Act 2023 Compliant):** Complies with the Digital Personal Data Protection Act 2023 by never storing raw Aadhaar digits. Generates a salted SHA-256 hash and masks the UI display to XXXX-XXXX-4821.
- **7C. Host Electricity Bill (Discom) Verification:** Eliminates fake listings by verifying the host's Consumer Account (CA) number against state power distribution utilities (BESCOM, TPDDL, MSEDCL), confirming legal possession of the physical premises.
- **7D. NPCI UPI ■1 Penny Drop Verification:** Validates the host's bank account title against their registered identity, guaranteeing zero failed payouts.

Feature 8: Objective Trust Index (OTI) Telemetry Engine

Objective Trust Index (OTI) Formulation

OTI Formulation: $OTI = (0.35 \times \text{Punctuality}) + (0.35 \times \text{Cleanliness}) + (0.20 \times \text{Identity Trust}) + (0.10 \times \text{Dispute History})$

- **Punctuality (35%):** Logged from GPS/QR timestamps. Includes a 10-min grace period; late checkout reduces punctuality score linearly.
- **Cleanliness Match (35%):** Direct algorithmic output of the Computer Vision Room Condition Delta evaluation.
- **Identity Trust (20%):** 100 points for verified DigiLocker/Aadhaar/Discom accounts; 70 points for unverified profiles.
- **Financial Record (10%):** 100 baseline; incurs a -15 point penalty for each formal dispute or unpaid overtime invoice.

Features 9 – 12: Concierge, Personas, Security & Schema

Feature 9 & 10: LoopBot AI Concierge & In-App Inquiry Dispatch

- **LoopBot AI Concierge:** Embedded multi-turn floating assistant assisting seekers with space recommendations, legal micro-lease explanations, and advising owners on staging and revenue maximization. Features conversational context memory and automatic heuristic fallback.
- **Direct Inquiry Subsystem:** Seekers can submit specific questions (e.g. ceiling clearance, 3-phase power). The AI engine pre-answers routine questions using verified metadata before notifying the host, resolving 70% of inquiries instantly.

Feature 11 & 12: Interactive Persona Switcher & Platform Security

- **Dual-Role Persona Switcher:** Instant switching between Seeker and Host views for rapid testing and live hackathon demonstrations. Includes a live in-room session console tracking remaining time, caretaker PIN, GPS distance, and checkout diff triggers.
- **Defense-in-Depth Security:** Enforces strict input validation, length capping, X-Content-Type-Options: nosniff headers, IP-based rate limiting on AI routes, and strict boundary delimiters preventing prompt injection attacks.

4. Relational Data Architecture & Core Models

Model Entity	Primary Schema Attributes	Architectural Responsibility
User (users)	id, name, email, password_hash, role, is_student_verified, is_aadhaar_verified, is_host_verified, objective_trust_score, discom_provider, upi_vpa_masked	Identity & trust profile. Stores DPDP-compliant masked credentials, NPCI UPI verification status, and calculated OTI scores.
Space (spaces)	id, owner_id, title, category, address, latitude, longitude, geofence_radius_meters, physical_access_type, keybox_code, room_qr_token, sqft, price_hourly, ai_lighting, ai_noise_level	Spatial catalog & property metadata. Stores AI-inspected environmental features, printable door QR tokens, and geofencing coordinates.
Booking (bookings)	id, space_id, renter_id, start_time, end_time, hours_booked, total_price, status, session_state, arrival_pin, checkin_gps_lat, condition_match_score, escrow_status, micro_lease_agreement	Transaction & telemetry record. Encapsulates synthesized AI micro-lease, session state transitions, GPS coordinates, and escrow refund status.
Review (reviews)	id, space_id, user_id, rating, comment, created_at	Subjective qualitative feedback supplementing the objective telemetry calculations.
SpaceInquiry (space_inquiries)	id, space_id, user_id, question, ai_answer, created_at	Host-guest inquiry logs with automated AI pre-answers.

5. Unit Economics, Competitive Edge & Hackathon Pitch

Market Disruption & Financial Dynamics: Traditional real estate sharing platforms operate with massive operational friction. **SpaceLoop** carves a completely new category: **Hyper-Local, Zero-Hardware Micro-Utility Rentals**. By eliminating hardware capital expenditure (CapEx) and statutory tenancy friction, SpaceLoop achieves unmatched unit economics:

Feature / Vector	Traditional Rental	Coworking Spaces	SpaceLoop Platform
Listing Onboarding	Manual measurement & broker	Central corporate facility	Instant AI Multimodal Scan
Contract Type	11-Month Tenancy Agreement	Monthly Desk Membership	Automated AI Micro-Lease (Sec 52)
Minimum Booking	11 Months to 3 Years	1 Full Day to 1 Month	1 Hour Flexible Granularity
Hardware & Access	Physical Keys & Lock Changes	RFID Cards & NFC Turnstiles	■0 Hardware (GPS + Door QR + PIN)
Identity & Trust	Paper police verification	ID photocopy on file	DigiLocker + OTI Telemetry

Unit Economics Breakdown & Host Yields

- **15% Platform Take-Rate:** High gross margin software business model with zero ongoing hardware maintenance overhead.
- **■0 Host CapEx:** Eliminates high upfront costs; a laminated ■10 printable door QR is the sole physical requirement.
- **Instant ■100 UPI Escrow Deposit:** Protects property owners against trash and power waste while keeping student renter friction near zero.
- **Democratic Urban Access:** Delivers quiet, air-conditioned study pods to students at ■25–■50/hour while unlocking ■12,000–■25,000/month of passive income for urban homeowners.

Hack2Ignite 2026 Winning Thesis

Hack2Ignite 2026 Pitch Summary: SpaceLoop proves that unused urban real estate is not an asset deficit, but a software distribution problem. By synthesizing Multimodal Spatial AI, India Stack Zero-Hardware Verification, and Automated Micro-Leasing, SpaceLoop creates a liquid, friction-free marketplace where every square foot has purpose, protection, and passive income potential.